VISION-LANGUAGE MODELS

Lecture 3

Bridging Vision and Language: Fusion Architectures

Language + Vision— what is missing?

- **Image side:**

  CLIP can MATCH text to images — it can't say anything fine-grained about image

  – One 512-d vector per image; answers limited to ranking candidate sentences you supply

  – E.g. "What is unusual about this image?" → has no candidate list → it needs open-ended text generation

- **Text side (assuming we have progressed from simple transformer to LLM):**

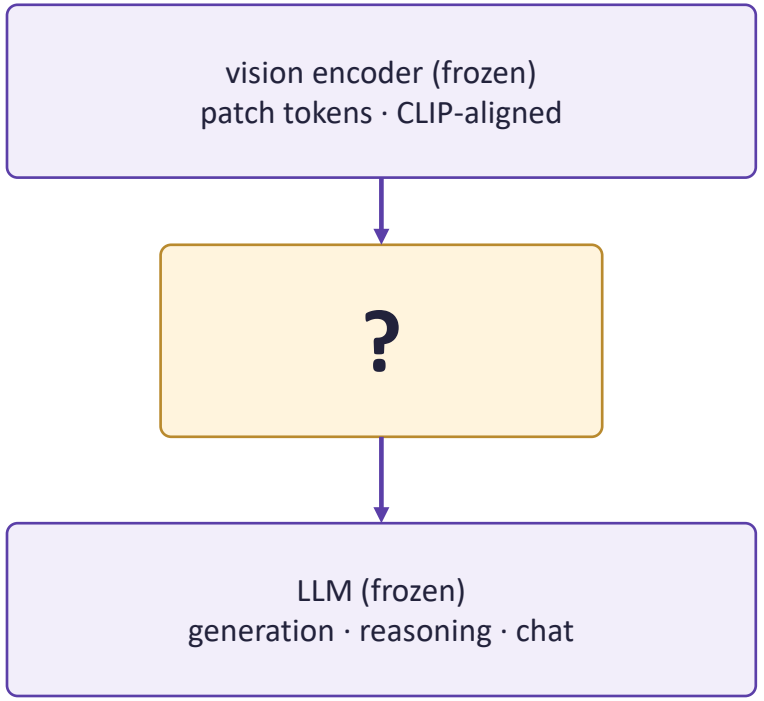
  LLMs generate, reason, follow instructions, but have no access to image understanding

- We have a vision encoder and an LLM, both expensive, both pre-trained and frozen (why?)

  How do you connect them so the LLM can condition its text generation on pixels?

- We need a Bridge:

A component that maps the visual embedding into a representation that looks like text embedding sequence that the LLM can parse and reason about.



Outline

I

Design constraints

Two axes: where does vision enter the LLM (cross-attention vs concatenation) and what gets trained? Also the token-budget problem.

II

Learned bridges

Flamingo: Perceiver Resampler + gated cross-attention into a frozen LM. BLIP-2: the Q-Former. 2022–23 state of the art.

III

Can simple solution beat complex approaches → LLaVA

A two-layer MLP and better data (visual instruction tuning) often beats sophisticated bridges. Why?

IV

Recent trends

Native multimodality: interleaved tokens, one transformer; where production VLMs (Qwen-VL, InternVL) are.

PART I

Design constraints

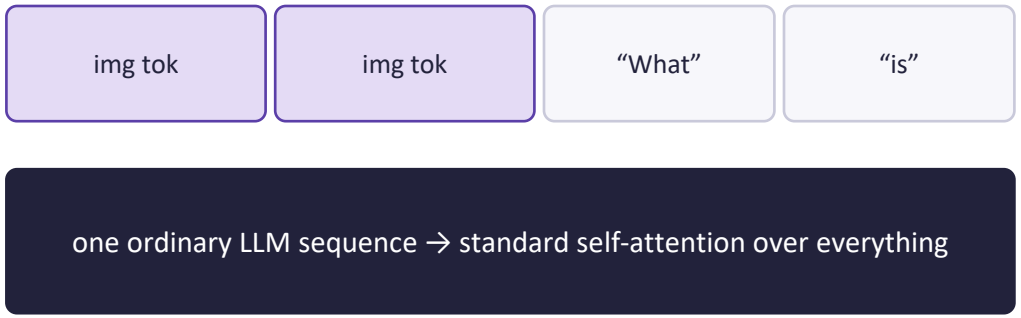
Components to optimize over

component	what it gives us	interface	cost to retrain from scratch
vision encoder (ViT, CLIP/SigLIP-pretrained)	patch tokens rich in semantics — already language-adjacent due to contrastive pretraining	$N \times D_v$ tokens, e.g. 576×1024 for ViT-L/14@336	high (See CLIP!)
LLM (GPT-style decoder)	generation, world knowledge, reasoning, instruction following	consumes sequences of D_t-dim token embeddings	very high (weeks × thousands of GPUs)
the bridge	maps vision tokens into something the LLM treats as input	$D_v \rightarrow D_t$, and possibly N → fewer	low — this is the part we can afford to train

- Cost of retraining vision+ text components is very high - connect them with the fewest new parameters possible
- CLIP-style pretraining of the vision tower is beneficial: its features already correlate with language
- The bridge has rich image features to utilize, not simply pixels

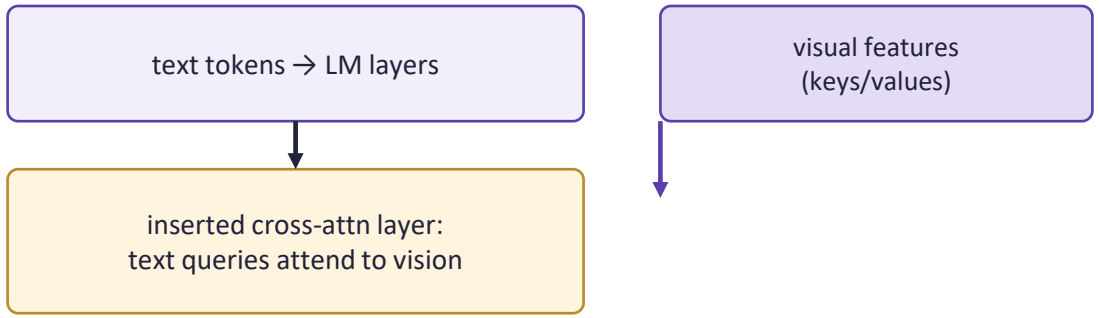
Design axis 1: Where is vision introduced?

Option A — concatenation ("visual tokens as words")



- LLM is unmodified — vision arrives as "foreign words" via a projection
- Note that we do NOT use the *single* image embedding that CLIP produces – instead we access the full set of ViT output-per-patch token embeddings *before* these are pooled for CLIP; these are 'img tok' referred to in figure.
- Every layer sees both ("aligned") vision and text tokens; simplest to implement (used in LLaVA, and most modern VLMs)
- Cost: image tokens occupy context — 576/image adds up rapidly

Option B — cross-attention ("vision on the side")



- New layers are added between frozen LM blocks; vision never enters the LLM token sequence
- Since LLM context not occupied by image tokens, more available for model reasoning — great for many images interleaved with text (Flamingo)
- Cost: new parameters at every insertion; more invasive changes to on the LM

Taxonomy after Alayrac et al. 2022; Liu et al. 2023.

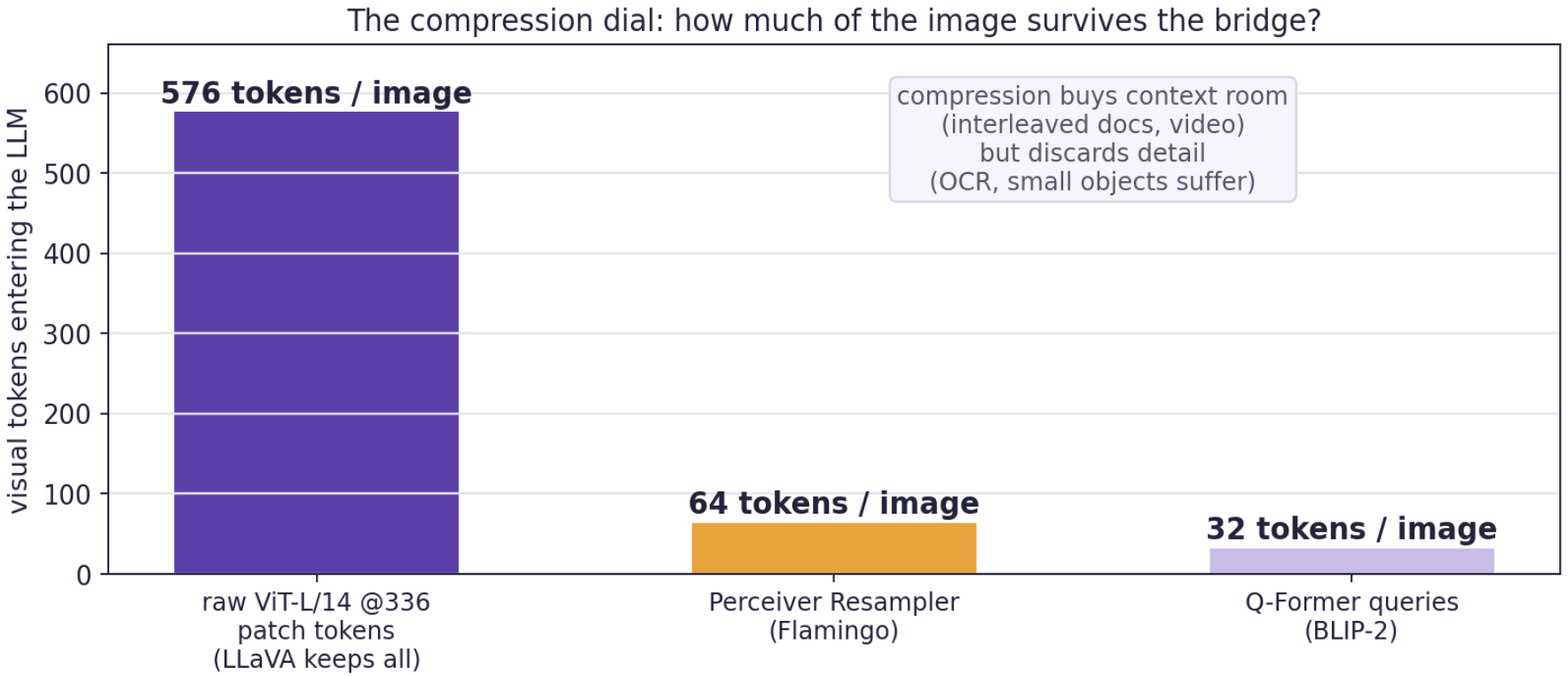
Design axis 2: What do you train?

Training Approach	vision encoder	bridge	LLM	trainable share	used by
bridge-only	**frozen**	**trained**	**frozen**	~1–3%	Flamingo, BLIP-2, LLaVA stage 1
bridge + LLM	**frozen**	**trained**	**fine-tuned**	95-98%	LLaVA stage 2
everything	**trained**	**trained**	**trained**	100%	native multimodal (Part IV)

- Why freezing was favored:
  - Compute: training 1–3% of parameters fits academic/research compute resources;
  - Context preservation: fine-tuning an LLM on small multimodal (text/image token) data risks catastrophic forgetting of its language ability
    - frozen = the LLM's reasoning skills retained
  - Data: web image-text pairs are plentiful
  - Data: high-quality *multimodal data* was scarce (until LLaVA created it – see later)

Alayrac et al. 2022 §2; Li et al. 2023 §1; Liu et al. 2023 §4.

The token-budget problem



- Motivation for compression – resource limited in ~2022
- **Example:** ViT-L/14 @336 → 576 patch tokens per image
- So, a webpage with 8 images ≈ 4,600 visual tokens
- This is before any text tokens are considered
- In a 2022 (with common context limits around 2k tokens for LM) this was not viable
- This motivated compressing input visual token stream with some learned model

- BUT by 2024+ things changed: contexts grew to million(s) tokens
- So keeping ALL tokens (or certainly many more) became viable — detailed (fine grained) image/text interaction was now possible

Token counts: Dosovitskiy et al. 2020; Alayrac et al. 2022; Li et al. 2023.

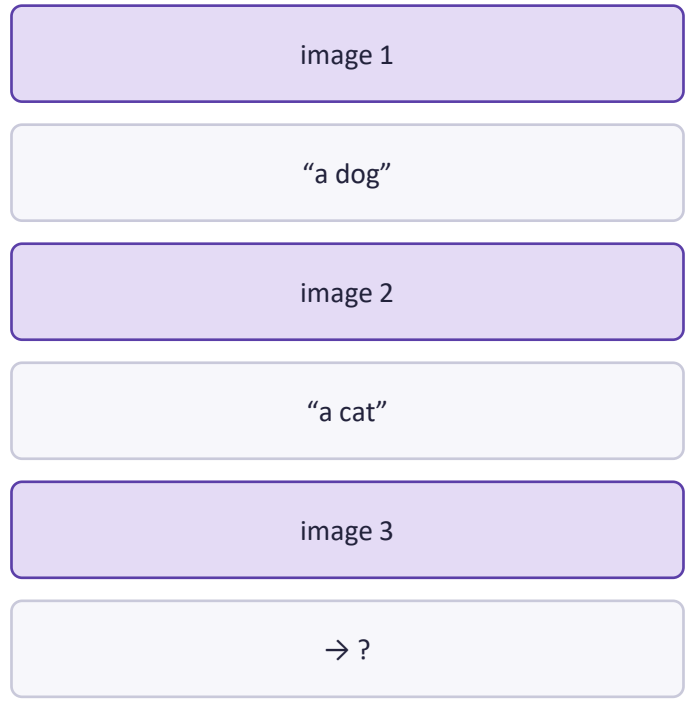
PART II

Learned bridges

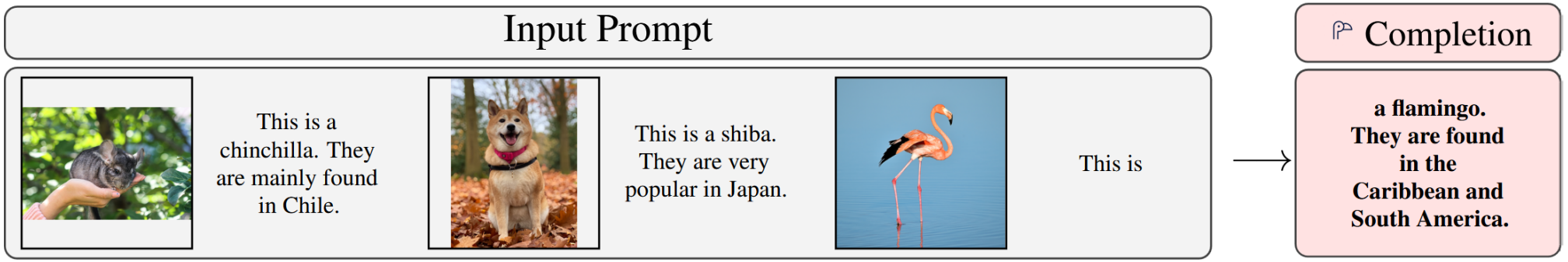
Flamingo and BLIP-2

Flamingo: few-shot learning + vision

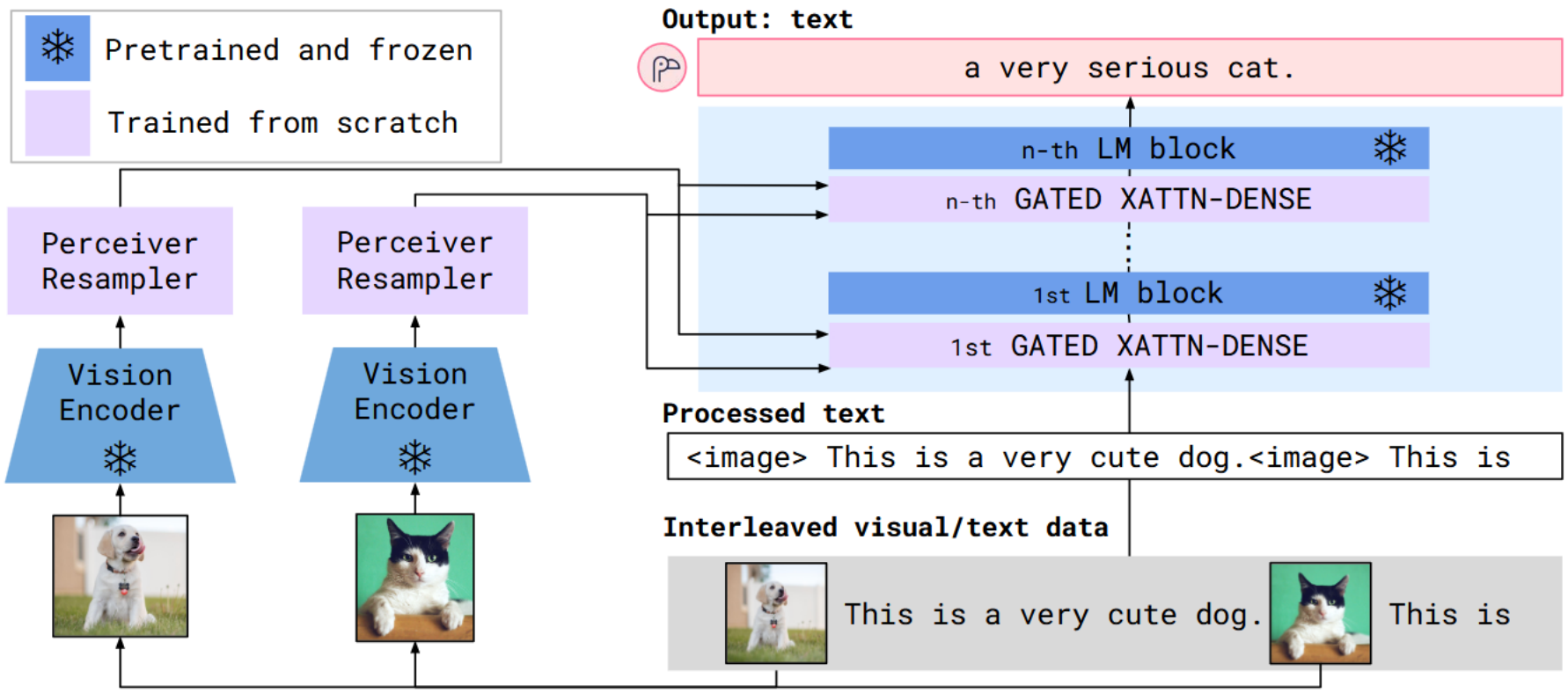
- **The idea: use** GPT-3-style few-shot prompting, but with images in the prompt too
  - Input is an interleaved sequence: image, text, image, text… — e.g. 4 (photo, caption) examples, then a 5th photo → model continues with its caption
  - So the design must (a) handle many images per prompt sequence and (b) not disturb the pretrained, frozen LLM (in this case Chinchilla LLM, up to 70B parameters)
- **Two new modules, both small relative to the size of LM:**
  - Perceiver Resampler — compress an image (or n video frame) features into 64 tokens
  - Gated cross-attention layers — inserted between frozen LM blocks; text attends to vision
- Result (in 2022): a single model, 32-shot-prompted, beat or matched fine-tuned task-specific SOTA on 6 of 16 benchmarks



interleaved few-shot prompt

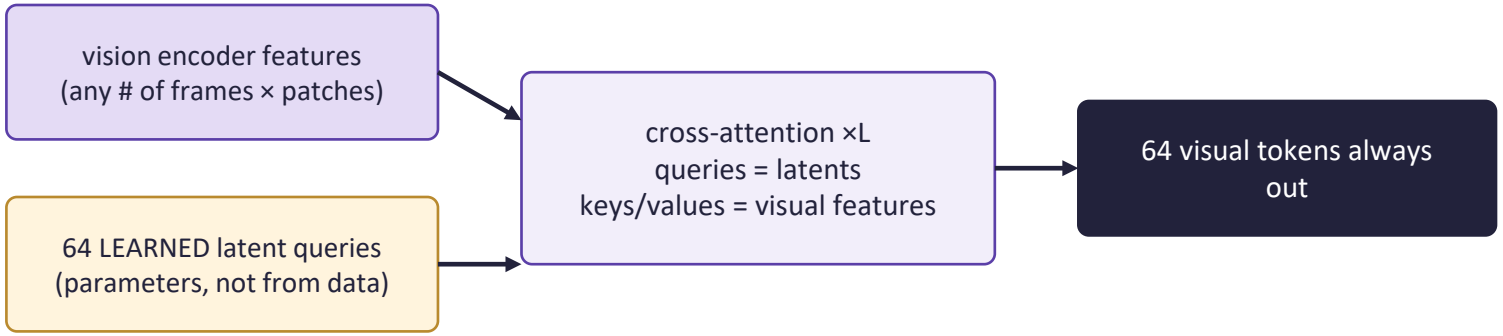


Flamingo: architecture

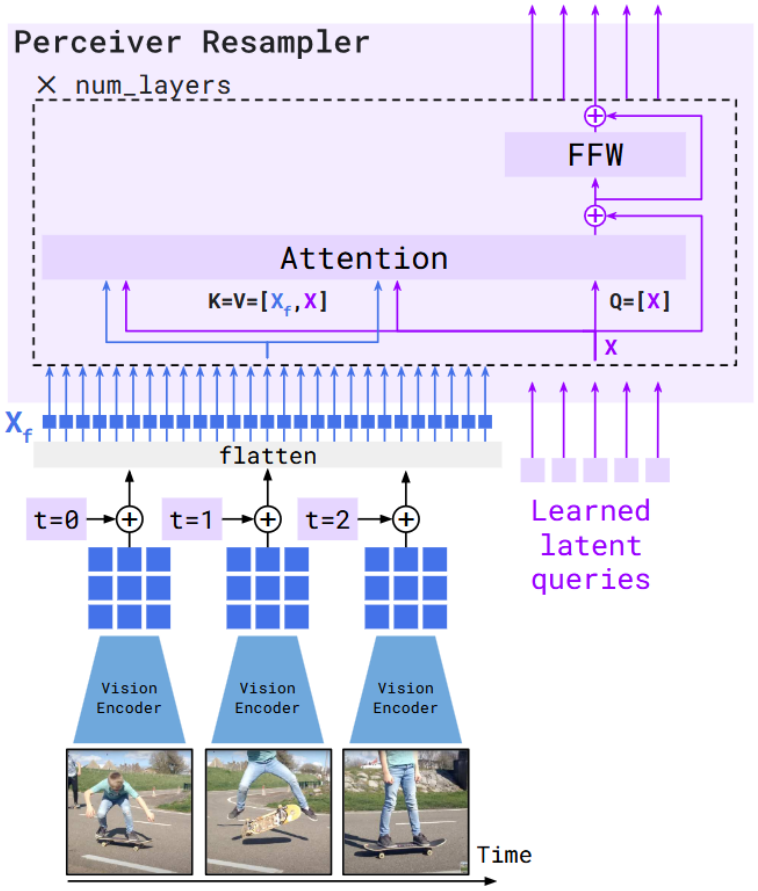


Module 1: the Perceiver Resampler

fixed-size output from variable-size input

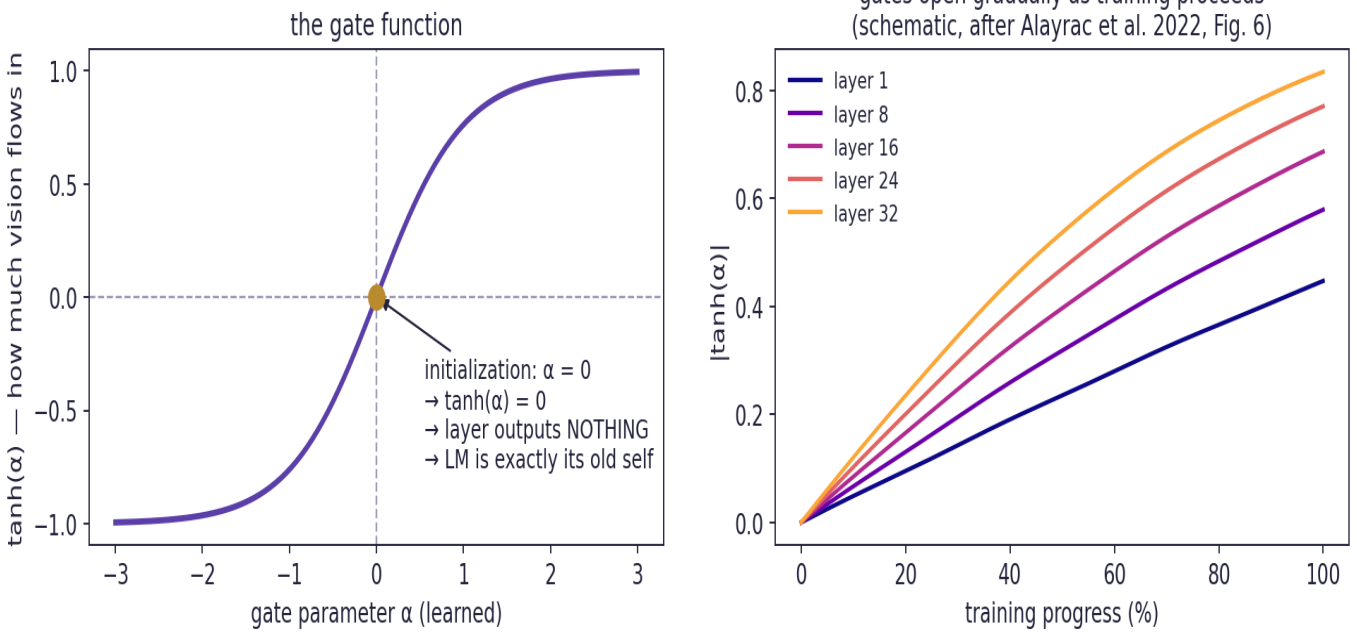


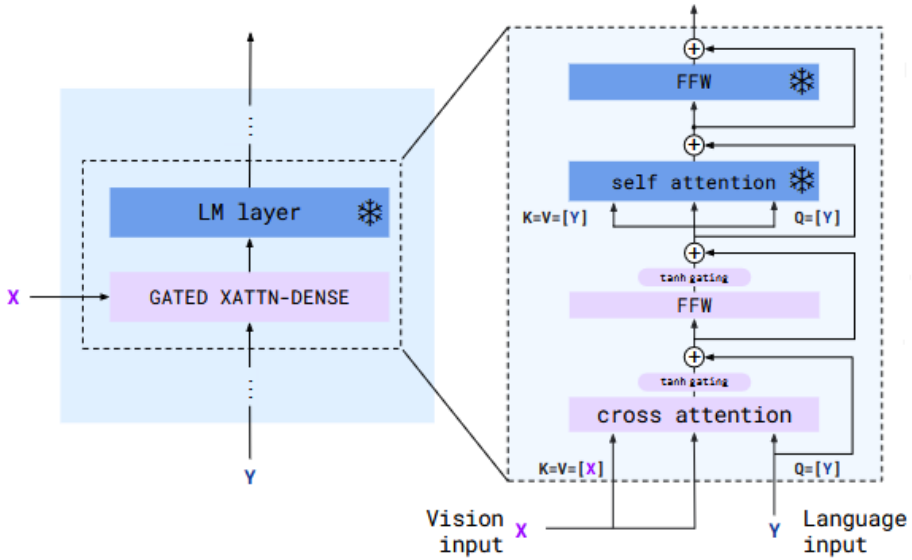
- **The trick: attention where the queries (Q) are learned parameters**
  - Output length = number of queries (64), regardless of input length — 1 image or 30 video frames in, 64 tokens out
  - Each query learns to "ask for" a kind of content (objects, layout, text, …) — a learned, fixed-size summary
  - Why Flamingo needs it: interleaved prompts contain many images/videos; 64 per image keeps cross-attention cheap and uniform
- The idea of "learned queries compressing a modality" reappears as BLIP-2's Q-Former and in video models - see later



Module 2: gated cross-attention

$$z \leftarrow z + \tanh(\alpha) \cdot \mathrm{XAttn}(\text{text}, \text{vision})$$
$$z \leftarrow z + \tanh(\alpha) \cdot \mathrm{FFW}(z)) \qquad \alpha \text{ init} = 0$$

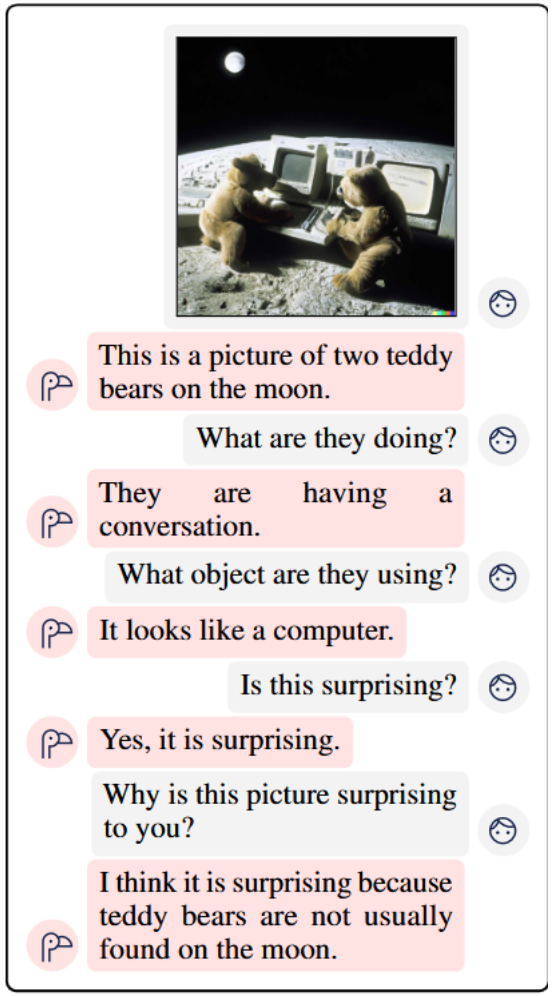




- At initialization tanh(0)=0: the new layers are invisible — the network is the original frozen LM, exactly. Training then opens each gate only as fast as it proves useful (right panel: deeper layers open differently).

Why the gating is needed

- **The problem with naive insertion**
  - Drop an untrained cross-attention layer into a 70B LM and its random outputs corrupt every downstream activation — the LM's language ability is damaged before learning starts
- **The gate as a guarantee**
  - tanh(α)·(new stuff) with α=0 → the added layers compute the identity at step 0
  - The model starts from "fully functional blind LM" and continuously shifts toward "vision enabled LM" — never passing through a "broken" state
  - Training signal stays clean: early gradients flow into the resampler/bridge while the LM's behavior is intact
- **A general principle found elsewhere**
  - "Initialize new additions to identity/zero" recurs across the field: zero-init projections, LoRA's zero-init B matrix, ControlNet's zero-convolutions
- **Observation:** gating solves a "training dynamics" problem - how to ensure stability and preservation of existing frozen knowledge as we integrate new data



Alayrac et al. 2022, §2.2; Hu et al. 2021 (LoRA, arXiv:2106.09685); Zhang et al. 2023 (ControlNet, arXiv:2302.05543).

Flamingo: summary

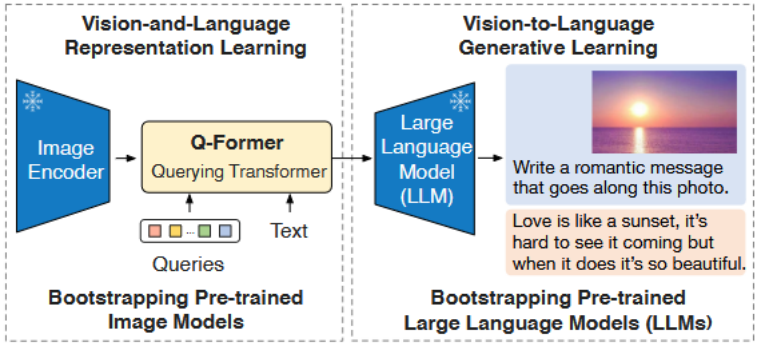
	Flamingo-80B (2022)
Language model	Chinchilla-70B — language skills inherited, frozen
vision model	contrastively-pretrained NFNet-F6 (their in-house CLIP-style encoder — note: **not** a ViT; the approach is in fact encoder-agnostic)
#new parameters	~10B (resampler + gated layers) on top of frozen 70B
training data	interleaved web pages (M3W) + image/video-text pairs — no human labels
This research showed	few-shot prompting beats fine-tuned SOTA on 6/16 tasks; strong VQA/captioning with 32 examples

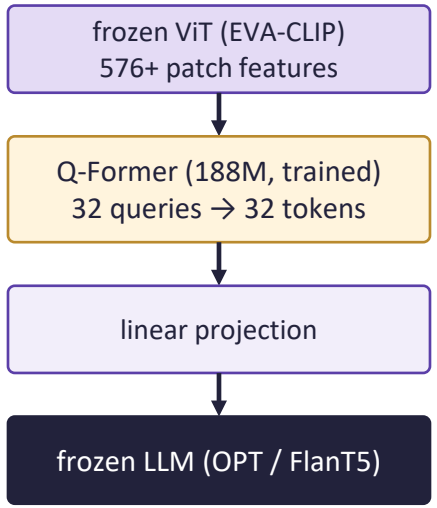
- **Limits and open-source alternatives:**
  - Closed weights + enormous → nobody outside DeepMind could build on it directly
  - Few-shot prompting ≠ conversation: it completes patterns, it doesn't follow instructions ("chat" needed a different ingredient — see Part III)
  - Paved way for open-source initiatives: OpenFlamingo and IDEFICS (HuggingFace) reproduced the recipe publicly in 2023

Alayrac et al. 2022, Tables 1–2; Awadalla et al. 2023 (OpenFlamingo, arXiv:2308.01390); Laurençon et al. 2023 (IDEFICS/OBELICS, arXiv:2306.16527).

BLIP-2: Bootstrapping Language-Image Pretraining

- **Approach: make the bridge itself a small pretrained model**
  - The Q-Former: a BERT-sized (~188M) transformer with 32 learned query tokens that extract language-relevant visual features
- **Bootstrapping:** we "bootstrap" good general image-language perf. from many examples
- **Two-stage approach to train Q-Former (Query Transformer):**
  - Stage 1
    - learns vision-language alignment against the frozen image encoder (optimizes 3 objectives, see later)
    - no LLM involved in this part
  - Stage 2
    - feed the 32 output tokens (via one linear layer) into a frozen LLM (OPT / FlanT5);
    - Further train the Q-Former to generate image conditioned prompts
    - Can be used with decoder or encoder-decoder LLMs
- **Intuition: the Q-Former interprets between two "language experts" — it learns the vision language first (stage 1), then how to phrase things for the LLM (stage 2)**

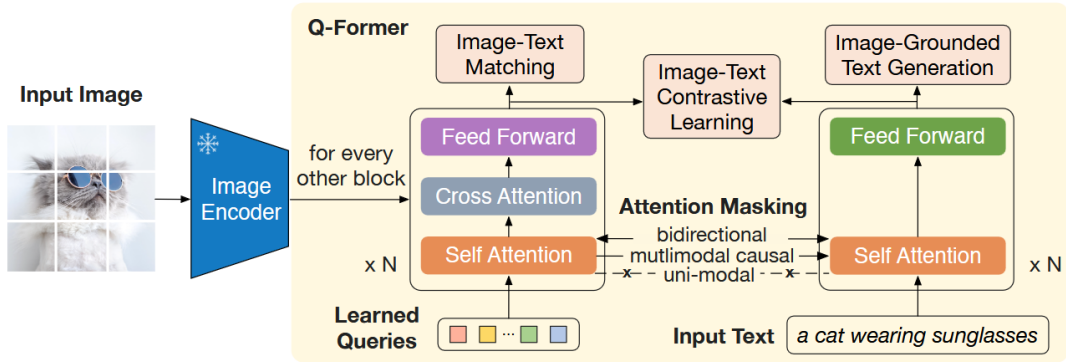




Li et al. 2023, "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models" (arXiv:2301.12597), §1–3.

Inside the Q-Former

- **Architecture: a BERT-base-initialized transformer processing two input streams**
  - Stream 1 — 32 learned query embeddings (768-d): interact with image features through cross-attention (xattn is inserted every other block)
  - Stream 2 — text tokens: share the same self-attention layers with the queries
  - Attention masks switch the mode: queries-only, queries↔text, or causal text (3 loss terms computed, jointly)
  - Output: exactly 32 tokens of 768-d — significant compression of the input visual token stream (vs say ViT)
- Similar idea to the Perceiver Resampler — learned queries as a fixed-size bottleneck — but the Q-Former's queries are also trained to be relevant to text, not just compact
- Assertion:
  - 32 well-chosen "visual sentences" suffice for most language tasks about an image.
- BUT not always… false for OCR/documents — detail is lost in the bottleneck.



Li et al. 2023, §3.1, Fig. 2.

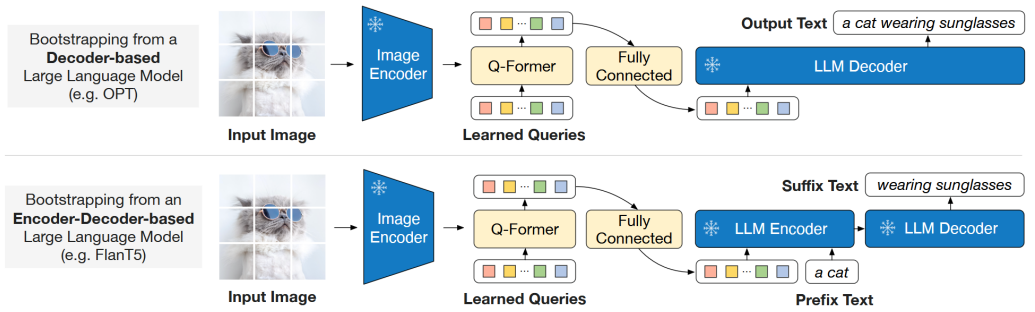
Pretraining

Stage 1 Pretraining

Loss term	task	attention mask
ITC — image-text contrastive	align query outputs with text embedding (best-of-32 similarity)	Image queries and text tokens only attend to themselves (unimodal)
ITM — image-text matching	binary classify: does this (image, text) pair match? (with hard negatives)	bidirectional: all queries attend to all text
ITG — image-grounded generation	generate the caption, conditioned on queries	causal on text; text sees queries

- Jointly optimized: contrastive gives global alignment, matching adds fine discrimination, generation forces completeness. The queries must satisfy all goals— that's what makes 32 tokens descriptive enough to be worth passing to an LLM.

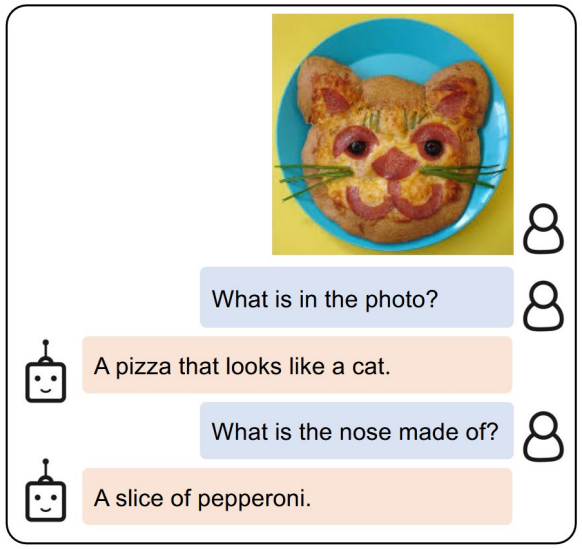
Stage 2 Pretraining

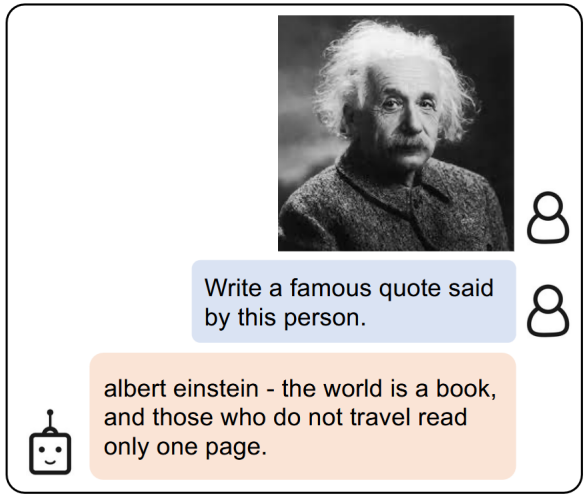


The Stage 1 pretrained Q-Former is connected to a MLP

- Projects to token dimension LLM expects (only 1-2M params)
- The Q-Former weights are adjusted again, along with MLP weights, to optimize for text generation criteria like “language modelling loss”
- No text enters the Q-Former in this stage – only the frozen LLM

BLIP-2: Summary





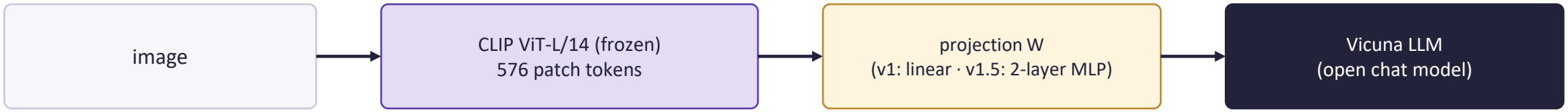
Inaccurate knowledge
(quote is from a different person)

- **Compared to Flamingo:**
  - 54x fewer trainable parameters
  - +8.7% performance on VQAv2 (visual question answering benchmark)
- **What this showed (early 2023):**
  - You do not need 10B new parameters to connect vision to language — a good 188M bridge suffices
  - Frozen instruction-tuned LLMs (FlanT5) pass their zero-shot instruction skills through the bridge — earlier expensive instructed-VQA training is retained
  - Greatly reduced compute: BLIP-2 trains in <9 days on 16×A100 — university scale. The "open VLM" wave (incl. LLaVA, InstructBLIP, MiniGPT-4) built directly on this
- **Where it still failed:**
  - Real conversation and instruction following remained weak — the LLM was frozen and the training data was captions, not dialogues
  - Only supports 1 image and text questions/prompts

PART III

LLaVA – a simple approach with lots of data

LLaVA: core architecture



"<image tokens> USER: What is unusual about this image? ASSISTANT: …" — vision enters as ordinary sequence tokens (axis-1 option A)

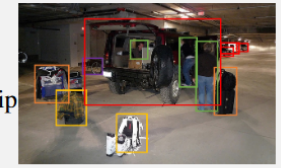
- **Compare the bridges:**
  - Flamingo added ~10B params of gated layers;
  - BLIP-2 pretrained, using a complex loss, a 188M Q-Former;
  - LLaVA simply trains a projection matrix
  - All 576 tokens are kept — no compression module at all: Fine image detail survives, but we sacrifice context to cover this detail
  - The LLM is open (Vicuna) and — crucially — will be fine-tuned in stage 2:
  - LLaVA abandons the "never touch the LM" doctrine
- **So where does the capability come from, if not the architecture? → The data.**

Liu et al. 2023, "Visual Instruction Tuning" (arXiv:2304.08485), §4; Liu et al. 2023b (LLaVA-1.5, arXiv:2310.03744).

The core innovation: manufactured instruction data

- **Date bottleneck ca. 2023:**
  - no dataset of (image, instruction, good response) triples existed
- **LLaVA's insight: have text-only GPT-4 write the dataset**
  - Feed GPT-4 the symbolic descriptions of a COCO image (captions + object boxes) — it never sees pixels
  - Prompt it to produce three types of output (158K samples in total):
    - conversations (58K) — multi-turn Q&A about the scene
    - detailed descriptions (23K) — dense paragraph captions
    - complex reasoning (77K) — questions needing inference ("what challenges might these people face?")

Context type 1: Captions
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes
person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>
Response type 2: detailed description
The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>
Response type 3: complex reasoning
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

- **This is knowledge distillation across modalities: a blind strong model teaches a sighted small one how to TALK about scenes. This pattern (synthetic instruction data) now underpins most open VLM training.**

Liu et al. 2023, §3, Table 1 (example adapted/abridged).

Two stage training

STAGE 1 · feature alignment
595K filtered image-text pairs (CC3M)
train: projection W only — ViT frozen, LLM frozen
“teach W to speak Vicuna's embedding dialect”

→

STAGE 2 · visual instruction tuning
158K GPT-4-generated instructions
train: W + the WHOLE LLM — ViT frozen
“now learn to converse about what you see”

- Why stage 1 exists: if W is random, LLM fine-tuning wastes its budget repairing garbled inputs — align the interface first, cheaply (only W has gradients)
- Why stage 2 fine-tunes the LLM (violating the frozen doctrine): instruction-following about images is a new skill; a frozen LM can imitate formats but not integrate vision deeply. Risk of forgetting is contained because stage-2 data is conversational (close to the LM's home distribution)
- Cost: LLaVA-1.5 trains in ~1 day on 8×A100 — the entire recipe is a university-lab budget. Compare: Flamingo (thousands of TPU-days), BLIP-2 (~16×A100×9d, still bridge-only)
- LLaVA-1.5's small deltas (MLP instead of linear, 336px input, academic VQA data mixed in) → SOTA on 11 benchmarks — architecture stayed trivial

Visual input example, Chicken Nugget Map:



User	Can you explain this meme in detail?
LLaVA	The meme in the image is a creative and humorous take on food, with a focus on chicken nuggets, as the center of the universe. The meme begins with the phrase "Sometimes I just look at pictures of the Earth from space and I marvel at how beautiful it all is..." The main image features a tray of cooked chicken nuggets, with the chicken pieces positioned to resemble the continents and islands on a world map. The meme playfully suggests that the chicken nuggets represent the Earth, and the various locations depicted in the photo are actually chicken nugget versions of different places. The punchline of the meme appears in smaller text at the bottom, reading "I mean, it's not the real Earth, but how beautiful it is all is." This lighthearted and entertaining meme combines food and humor, using the image of chicken nuggets to create a funny and imaginative representation of the world.
User	Can you explain this meme?
GPT-4 [36]	This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.
User	Can you explain this meme in detail?
BLIP-2	sometimes i just look at pictures of the earth from space and marvel how beautiful it is
User	Can you explain this meme in detail?
OpenFlamingo	It's a picture of a chicken nugget on the International Space Station.

Liu et al. 2023, §4–5; Liu et al. 2023b (LLaVA-1.5), §3.

Why did the simplest bridge win?

- **1 · The core components for language and vision had already been extensively trained on large data; little effort needed to bridge**
  - CLIP's contrastive pretraining aligned visual features to language; Vicuna's instruction tuning made the LM conversation-capable. The difference between them was small enough for an MLP
- **2 · Data quality beats complex architectural design**
  - 158K “on-distribution” *instruction dialogues* taught reasoning/conversational behaviour
  - no amount of caption data could accomplish this: BLIP-2's architecture with caption-style data still couldn’t “chat”
- **3 · Fine-tuning the LLM simplified integration**
  - Frozen-LM designs limit how deeply vision can influence reasoning; fine-tuning the LM (safely, on conversational data) removes that limit
- **4 · Simple model architecture was easy to extend**
  - Fewer integrated components → easy experimental reproduction, ablation, extension.
  - The research community iterated on LLaVA rapidly while Flamingo couldn't be iterated on at all
- **Design lesson: when pretrained (more sophisticated) expensive-to-train components are already capable, the interfacing components can be simple — so focus on getting more/better data (for fine-tuning etc) and well engineered loss functions instead**

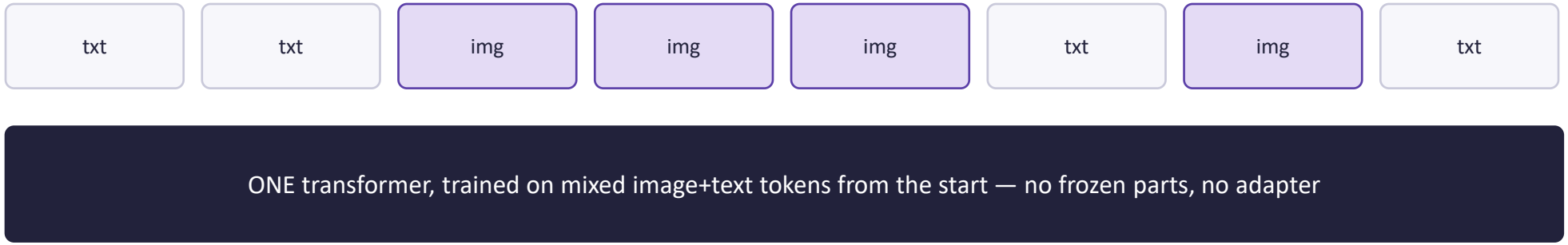
The three bridges compared

	Flamingo (2022)	BLIP-2 (2023)	LLaVA / 1.5 (2023)
bridge	Perceiver Resampler + gated cross-attn layers	Q-Former (BERT-sized, 3-objective pretrained)	linear → 2-layer MLP projection
where vision enters	cross-attention, every few LM blocks	as 32 prompt tokens (concatenation)	as 576 sequence tokens (concatenation)
tokens / image	64	32	576 — detail preserved
LLM	frozen (Chinchilla-70B)	frozen (OPT / FlanT5)	fine-tuned (Vicuna) in stage 2
Principal data used	interleaved web pages (M3W)	web image-text pairs	GPT-4-written instruction dialogues
Good	few-shot, multi-image, video-in-prompt	parameter efficiency (188M trained)	conversation & instruction following
Bad	closed, huge, no chat	bottleneck destroys fine detail; weak dialogue	context cost; single image (v1)

PART IV

Recent trends

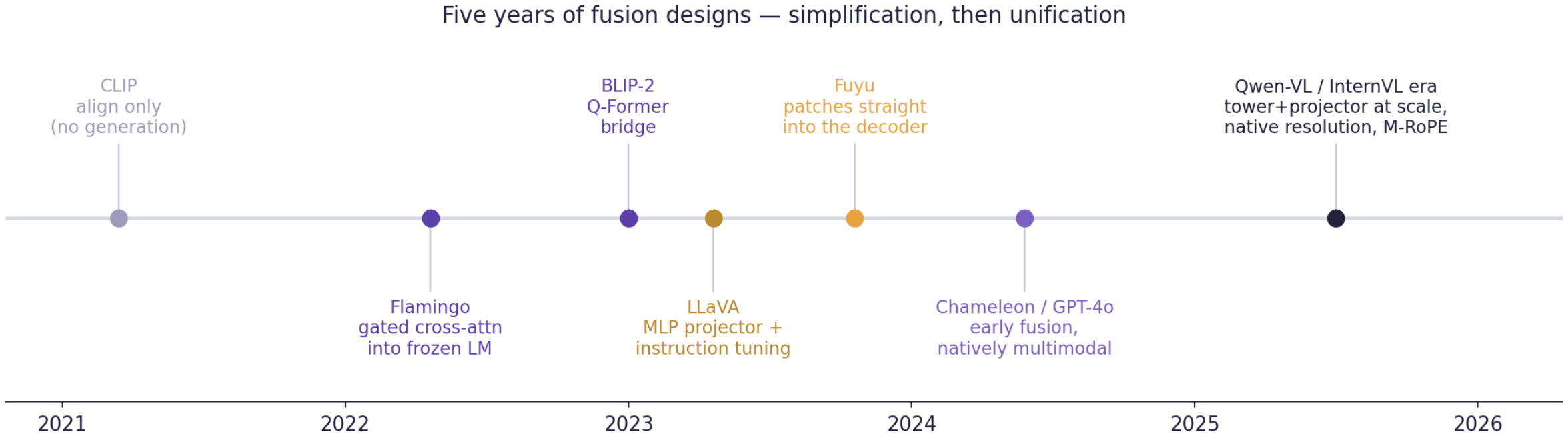
Native multimodality: remove the bridge entirely



- **Early fusion**: image and text are merged into one token stream before the transformer sees them (vs late fusion which uses existing towers to produce token for text or image).
- **Two kinds of "image tokens" considered:**
  - Continuous: patch embeddings straight into the decoder — Fuyu even deletes the complex vision encoder (a linear layer on raw patches becomes the "vision model"; recall 1st lecture - ViT's first layer is also just a projection)
  - Discrete: a learned tokenizer (a Vector Quantizer) maps images to codebook ids in the same vocabulary as text — Chameleon; enables generating images by the same next-token loop
  - Benefit: one set of weights combines both kinds of token — any token interleaving, understanding and generation supported in one model
  - Cost: full-scale pretraining from scratch (the axis-2 "everything" row) + real training-stability issues (Chameleon paper covers many tricks to try and overcome this)

Bavishi et al. 2023 (Fuyu-8B, Adept blog); Chameleon Team 2024 (arXiv:2405.09818).

2021 → 2026: how the field evolved



- **Observation**: production open VLMs (Qwen-VL, InternVL) did not go completely early-fusion. They scaled up the LLaVA shape: strong ViT tower + MLP projector + fine-tuned LLM — with native resolution, M-RoPE, and massive joint training
- Compression modules reduced/vanished for images (contexts grew; and image detail matters) — however, resampler idea still needed for video, where token counts explode again
- Early fusion dominates the unified understanding and generation frontier — we return to it later

Bai et al. 2025 (Qwen2.5-VL, arXiv:2502.13923); Zhu et al. 2025 (InternVL3, arXiv:2504.10479).

Which design for which job?

If you need…	choose…	because…
search / retrieval / deduplication at scale	two-tower CLIP/SigLIP	embeddings precompute & index; no fusion needed
chat about single images, fine detail (OCR, docs)	projector + instruction-tuned LLM (LLaVA type)	all tokens kept; conversation from instruction data
many images / video in one prompt, compact context	resampler/Q-Former-style compression (or cross-attn)	fixed small token cost per image
cheapest possible adaptation of existing models	frozen everything + small bridge (BLIP-2 type)	~1% trainable params; days of compute

Summary

Design space

Two axes: where vision enters (cross-attention vs concatenated tokens) & what trains (bridge only → everything). Effectively use of Tokens-per-image budget is core design priority.

Learned bridges

Flamingo: 64-token resampler + tanh-gated cross-attention (init 0 = preserve existing info) into a frozen LM. BLIP-2: a 188M Q-Former, 3-objective pretrained — 54× fewer trainable params than Flamingo, better zero-shot VQA.

LLaVA's lesson

An MLP + 158K GPT-4-written instruction dialogues + (carefully) fine-tuning the LLM beat the complex bridges. These are well designed components that benefit from much more data.

Current approach

Production VLMs scaled up the LLaVA approach (tower + projector, native resolution); early fusion dominates. Compression removed for images, but still needed for video.

Resources

- **Paper references/report**
  - Alayrac et al. (2022). Flamingo: a Visual Language Model for Few-Shot Learning. NeurIPS 2022. arXiv:2204.14198
  - Li et al. (2023). BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and LLMs. ICML 2023. arXiv:2301.12597
  - Liu et al. (2023). Visual Instruction Tuning (LLaVA). NeurIPS 2023. arXiv:2304.08485 · Liu et al. (2023b). Improved Baselines with Visual Instruction Tuning (LLaVA-1.5). arXiv:2310.03744
  - Chameleon Team (2024). Chameleon: Mixed-Modal Early-Fusion Foundation Models. arXiv:2405.09818 · Bavishi et al. (2023). Fuyu-8B (Adept technical blog)
  - Bai et al. (2025). Qwen2.5-VL Technical Report. arXiv:2502.13923 · Zhu et al. (2025). InternVL3. arXiv:2504.10479
  - Awadalla et al. (2023). OpenFlamingo. arXiv:2308.01390 · Dai et al. (2023). InstructBLIP. arXiv:2305.06500 · Hu et al. (2021). LoRA. arXiv:2106.09685
- **Useful blogs**
  - https://towardsdatascience.com/visual-question-answering-with-frozen-large-language-models-353d42791054/
  - https://towardsdatascience.com/flamingo-intuitively-and-exhaustively-explained-bf745611238b/
- **Code to try:**
  - Run LLaVA-1.5 or BLIP-2 from the HuggingFace (llava-hf/llava-1.5-7b-hf ; Salesforce/blip2-opt-2.7b)